RAGHAV SAREEN

(650) 660-7750 ◇ Pleasanton, CA
raghavsareen2006@gmail.com ◇ linkedin.com/in/raghavsareen2006 ◇ github.com/RaghavSareen2006

OBJECTIVE

Computer Engineering student pursuing Firmware and AI Engineering opportunities, with experience spanning embedded C/C++, STM32 motor control, edge-AI computer vision, LLM evaluation, agent testing, Linux systems, and hardware-software integration.

EDUCATION

Bachelor of Science in Computer Engineering, San Jose State University — Expected May 2028
GPA: 3.776/4.000
Selected Coursework: Programming Concepts, Object-Oriented Programming, Algorithms and Data Structures, Assembly Language Programming, Electronics, Circuit Analysis, Introductory Electrical Engineering Lab, Engineering Statistics, Differential Equations and Linear Algebra

SKILLS

Programming	C, C++, Python
Firmware & Embedded	STM32F103, libHAL, GPIO, CAN, PWM, I^2C, PID control, BLDC motors, limit switches
AI & Computer Vision	Ultralytics YOLO, TensorRT, OpenCV, prompt engineering, LLM evaluation, agentic workflows
Systems & DevOps	NVIDIA Jetson Orin Nano, ARM64 Linux, Docker Compose, Tailscale, Git/GitHub
AI & Product Tools	OpenClaw, Claude Code, ChatGPT, Jira, Kanban, Scrum, PRDs, epics, features, user stories

TECHNICAL EXPERIENCE

Firmware Team Member, ALICANTO Rover — Sep 2024 – May 2026
SJSU Robotics — *San Jose, CA*

- Programmed and tested C++ firmware for the ALICANTO rover arm subsystem in preparation for the 2027 University Rover Challenge; committed changes that were reviewed and merged into the shared team codebase.
- Validated three STM32F103 MicroMod and motor assemblies controlling the wrist, shoulder, and elbow, verifying motor response and component behavior with libHAL.
- Updated and tuned PID control functions to reduce arm jitter and improve stability during movement.
- Collaborated with the firmware lead and sublead to implement arm homing for BLDC motors using one limit switch and hands-on GPIO, CAN, and PWM communication.

AI Agent Testing Volunteer — Jun 2026
Social Experiments — *Remote*

- Designed 60 functional test cases and executed 37 functional tests plus 13 end-to-end workflow scenarios for an AI Slack agent.
- Tested Jira integration, contract drafting, document ingestion, redlining, approvals, role permissions, duplicate handling, error recovery, and response-time behavior.
- Built a structured test workbook containing priorities, preconditions, prompts, test steps, expected and actual results, automation candidates, and defect notes.
- Identified failures or errors in 28 scenarios, including duplicate responses, long timeouts, incorrect task routing, missing attachments, and incomplete workflow transitions.
- Reported results to the product team, contributing to improvements across most identified issues; used Jira and Agile practices and learned to structure PRDs with epics, features, and user stories.

PROJECTS

AI-Assisted Home Security Arm Turret Jul 2026 – Present
NVIDIA Jetson Orin Nano, xArm, YOLO, TensorRT, OpenCV, Docker Compose, Tailscale

- Conceived and assembled a customizable home-security system by integrating a USB camera and a commercial xArm kit with a Jetson Orin Nano; configured two servo motors for pan and tilt control.
- Used OpenClaw and Claude Code for AI-assisted implementation, then personally deployed, configured, troubleshot, and validated the complete hardware-software system for continuous 24/7 operation at home.
- Tested person and face detection, target verification and reacquisition, automatic room scanning, dog tracking, manual hold, digital zoom, photo capture, Telegram alerts, GPIO status lights, and safe servo limits.
- Deployed GPU inference and web services with Docker Compose and built private remote access through Tailscale; validated an authenticated dashboard with livestream controls and real-time telemetry.

Jetson Home Lab and Personal AI Infrastructure Jul 2026 – Present
Jetson Orin Nano, ARM64 Linux, OpenClaw, Docker Compose, Tailscale, Home Assistant

- Configured the Jetson as a multi-service home server running OpenClaw, Home Assistant, Music Assistant, Jellyfin, Portainer, and computer-vision containers.
- Managed persistent volumes, host networking, container health, logs, service restarts, and private HTTPS access through Tailscale.
- Diagnosed IoT connection failures, camera conflicts, container crash loops, high system load, swap pressure, and USB HID availability; configured exit-node forwarding and persistent NAT/firewall behavior.
- Created a sanitized Home Assistant configuration repository with Docker Compose files, restore documentation, secret exclusions, and credential-pattern checks.

The Efficacy of ChatGPT in Mathematics May 2023 – Sep 2023
Polygence Research Program

- Designed and conducted an experiment evaluating ChatGPT across 107 algebra problems from a standardized textbook dataset.
- Measured 81% initial answer accuracy, compared two follow-up strategies, and analyzed failure patterns involving graphs, rational expressions, and mathematical formatting.
- Evaluated LaTeX formatting and math-teacher role prompting as mitigation strategies; authored a research paper and presentation with guidance from a research mentor.
- Won Top Asynchronous Pitch among more than 300 presentations at the Polygence Symposium of Rising Scholars.

LEADERSHIP & ACTIVITIES

Digital Content Creator Jun 2020 – Present
Built and managed a pet-focused social media brand across Instagram, YouTube, and TikTok, reaching 30,000+ Instagram followers, 1,700+ YouTube subscribers, and 1.6M+ combined views through content planning, production, and audience analysis. Also produced personal-vlog content generating 140,000+ combined views across the three platforms.

Social Media Coordinator, Indian Students Organization Aug 2025 – Present
Created two promotional reels, participated in four to five event videos, attended weekly planning meetings, and supported promotion and event operations for a program that sold 75 tickets.

Program Member, MESA Engineering Program Sep 2024 – Present
Volunteered 12+ hours as an event floater for Science Extravaganza, supporting hands-on STEM workshops serving more than 350 middle-school students.

AWARDS & CERTIFICATIONS

Awards: 2026 President's Scholar, San Jose State University; Top Asynchronous Pitch, Polygence Symposium of Rising Scholars; AP Scholar; California State Seal of Biliteracy
Certifications: Command Line in Linux and Getting Started with Linux Terminal (Coursera, 2026); Learn Git and GitHub in One Day (Udemy, 2026); Claude Code in Action (Anthropic, 2026)

LANGUAGES

English (Full Professional Proficiency) ◇ Hindi (Native or Bilingual Proficiency) ◇ Spanish (Limited Working Proficiency)